

# Test Run Report — Run #119

Environment: Production • Trigger: Scheduled • Started: Aug 6, 2026 22:00 • Ended: Aug 7, 2026 00:01  
Generated: 8/7/2026, 7:28:43 AM

## Summary

|                                       |                                                       |                                                |                                               |
|---------------------------------------|-------------------------------------------------------|------------------------------------------------|-----------------------------------------------|
| <div>210</div> <div>Total Tests</div> | <div>88.4%</div> <div>Pass Rate (excl. skipped)</div> | <div>167 / 22</div> <div>Passed / Failed</div> | <div>21</div> <div>Skipped (connection)</div> |
|---------------------------------------|-------------------------------------------------------|------------------------------------------------|-----------------------------------------------|

## Failures (22)

| Failure Cluster            | Category            | Failures |
|----------------------------|---------------------|----------|
| Tool Execution Failures    | Tool Execution      | 10       |
| LLM Reasoning Issues       | Llm Reasoning       | 9        |
| Data Availability Issues   | Data Availability   | 2        |
| Graph/Visualization Errors | Graph Visualization | 1        |

### Tool Execution Failures TOOL EXECUTION (10 failures)

## 1. MOF

Production • 38.4s

**Question:** Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

**Error reason:** AI indicated failure (matched 'internal error'): Here's the summary for the requested clipping-loss analysis over the last six months: - Data Calculation: • The clipping-loss tool encountered an internal error (`unexpected keyword argument 'react`

## TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well",
    "displayed_response": "Here's the summary for the requested clipping-loss analysis over the last six months:\n\n- Data Calculation:\n  · The clipping-loss tool encountered an internal error (`unexpected keyword argument 'reactive_col'`) while processing this plant/time range.\n  · As a result, no clipping-loss values were returned for any of the 8 inverters.\n\n- Visualization:\n  · A chart titled “Clipping Loss per Inverter - Last 6 Months” has been prepared to display the clipping loss by inverter once data becomes available.",
    "coding_response": [
      "I couldn't calculate the clipping loss because the clipping-loss tool is currently failing internally for this plant/time range with the error: `unexpected keyword argument 'reactive_col'`. No clipping values were returned for the 8 inverters.",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping Loss per Inverter - Last 6 Months",
        "yAxisTitle": "Clipping Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "functions.calculate_clipping_or_curtailment_losses_ag",
      "functions.inspect_xpool"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from February 6, 2026, to August 6, 2026, including the last day.\n2. I find all relevant inverter power columns in the dataset, making sure to leave out any columns related to reactive power or flags.\n3. I check that the required power columns are actually present in the data.\n4. I clean up the plane of array (POA) irradiance data by filling in any missing values using a method that accounts for daily sunlight patterns.\n5. For each inverter power column, I calculate the power loss by taking the absolute difference between the measured power and the inverter's rated AC power during periods when losses are detected.\n  - I identify clipping events using both a machine learning approach (rdtools XGBoost clip filter) and a simple rule: if the power is above 85% of the inverter's rated AC power, it's considered clipped.\n6. I convert the detected power losses into energy lost (in kWh) by multiplying the loss by the time interval (0.0833 hours).\n7. I add up all the energy losses within the selected time period to get a total loss value for each inverter.\n8. I organize the results into a DataFrame, with each inverter as a column and each aggregation period (like day or month) as a row.\n9. I return the results as a dictionary, where each key is an aggregation period (or just a single total row), depending on the settings."
  }
}
```



Here's the summary for the requested clipping-loss analysis over the last six months:

- Data Calculation:
  - The clipping-loss tool encountered an internal error ( " unexpected keyword argument 'reactive\_col' ") while processing this plant/time range.
  - As a result, no clipping-loss values were returned for any of the 8 inverters.

- Visualization:
  - A chart titled "Clipping Loss per Inverter – Last 6 Months" has been prepared to display the clipping loss by inverter once data becomes available.



10:49 AM



Are you satisfied with the answer?



Time taken: 23s, token: 0 tokens

connected to Mof



How can we help you today?



LEAVE ANY  
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

## 2. MOF

Production • 38.4s

**Question:** Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

**Error reason:** AI indicated failure (matched 'internal error'): Here is the summary of your request: • **Data Availability:** - Clipping-loss tool encountered an internal error (unexpected-argument error) and could not produce clipping-loss values by inverter for t

### TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well",
    "displayed_response": "Here is the summary of your request:\n\n• Data Availability:\n - Clipping-loss tool encountered an internal error (unexpected-argument error) and could not produce clipping-loss values by inverter for the last two months.\n\n• Visualization:\n - The visualization has been prepared: a line chart titled \"Clipping Loss per Inverter Over the Last 2 Months.\" ",
    "coding_response": [
      "I couldn't calculate the clipping loss because the clipping-loss tool is currently failing internally for this plant/query (it returns an unexpected-argument error before producing any inverter values).",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping Loss per Inverter Over the Last 2 Months",
        "yAxisTitle": "Clipping Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "functions.calculate_clipping_or_curtailment_losses_ag",
      "functions.inspect_xpool"
    ],
    "urls": [],
    "steps": "1. Filter the data to include only records from June 6, 2026, to August 6, 2026, including the end date.\n2. Identify all relevant inverter power columns in the dataset, making sure to exclude any columns related to reactive power or flags.\n3. Check that the required power columns are present in the data.\n4. Clean the plane of array (POA) irradiance data by filling in any missing values using a method that accounts for daily sunlight patterns.\n5. For each inverter power column, calculate the power loss by finding the absolute difference between the measured power and the inverter's rated AC power during periods of detected loss.\n - Detect clipping events using both a machine learning approach (rdtools XGBoost clip filter) and a threshold method, flagging any power readings above 85% of the inverter's rated AC power.\n6. Convert the detected power losses into energy lost (in kWh) by multiplying by the time interval (0.0833 hours).\n7. Summarize the energy loss for each inverter, either by day, month, or another chosen time period.\n8. Provide a detailed breakdown of loss values for each inverter and each time period.\n9. Organize the results into a DataFrame, with inverters as columns and time periods as rows.\n10. Return the results as a dictionary, either with detailed breakdowns by period or as a single total, depending on the chosen settings."
  }
}
```

Return the last 2 months of inverter data, list them down and plot it as well



Here is the summary of your request:

- Data Availability:
  - Clipping-loss tool encountered an internal error (unexpected-argument error) and could not produce clipping-loss values by inverter for the last two months.
- Visualization:
  - The visualization has been prepared: a line chart titled "Clipping Loss per Inverter Over the Last 2 Months."

10:56 AM



Are you satisfied with the answer?



Time taken: 23s, token: 0 tokens

● connected to Mof



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LEAVE ANY  
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

### 3. KUPCO\_HQ\_DAJEEJ

Production • 55.2s

**Question:** Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

**Error reason:** AI indicated failure (matched 'internal error'): Summary of Clipping Loss Analysis (Last 6 Months) Data Quality Notes: - The clipping-loss tool encountered an internal error ('reactive\_col' argument error) when processing the dataset. - No clipping

## TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well",
    "displayed_response": "Summary of Clipping Loss Analysis (Last 6 Months)\n\nData Quality Notes:\n- The clipping-loss tool encountered an internal error (`reactive_col` argument error) when processing the dataset.\n- No clipping-loss values could be calculated or returned.\n\nAttempt Details:\n- Time window: 2026-02-06 to 2026-08-06 (last 6 months)\n- Inverters evaluated:\n  - inverter1\n  - inverter2\n  - inverter3\n- Result: No reliable per-inverter clipping-loss breakdown available due to calculation failure.\n\nVisualization:\n- A chart illustrating the clipping loss for each inverter over the requested period has been created.",
    "coding_response": [
      "I couldn't calculate clipping losses because the clipping-loss tool is currently failing internally for this dataset: it returns an unexpected `reactive_col` argument error even when called with valid clipping parameters.\n\nI attempted the last 6 months window from 2026-02-06 to 2026-08-06 for all inverters:\n- inverter1\n- inverter2\n- inverter3\n\nBut no clipping-loss values were returned, so I can't provide a reliable per-inverter breakdown.",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping loss for each inverter - last 6 months",
        "yAxisTitle": "Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "functions.inspect_xpool",
      "functions.total_energy_func_ag",
      "functions.calculate_clipping_or_curtailment_losses_ag"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from February 6, 2026, to August 6, 2026, including the end date.\n2. I select the relevant inverter power columns (for example: inverter1_active_power, inverter2_active_power, inverter3_active_power).\n3. I check that these power columns are present in the dataset.\n4. I clean the plane of array (POA) irradiance data by filling in any missing values using a method that accounts for daily sunlight patterns.\n5. For each inverter, I calculate power losses by finding the difference between the measured power and the inverter's rated AC power during periods of loss. I identify these loss periods (clipping events) using both a machine learning filter (rdtools XGBoost clip filter) and a rule that flags power above 85% of the rated AC power.\n6. I convert these power losses into energy losses (in kWh) by multiplying the loss by the time interval (0.0833 hours).\n7. I group or sum the energy loss values by the chosen time period (such as by day, month, or the entire period) for each inverter.\n8. I organize the results into a DataFrame, with each inverter as a column and each time period as a row.\n9. I return the results as a dictionary, where each key is a time period (or a single total), and the values show the losses for each inverter."
  }
}
```



- no clipping-loss values could be calculated or returned.

Attempt Details:

- Time window: 2026-02-06 to 2026-08-06 (last 6 months)
- Inverters evaluated:
  - inverter1
  - inverter2
  - inverter3
- Result: No reliable per-inverter clipping-loss breakdown available due to calculation failure.

Visualization:

- A chart illustrating the clipping loss for each inverter over the requested period has been created.

11:06 AM

Are you satisfied with the answer?



Time taken: 39s, token: 0 tokens

connected to Kupco Hq Dajeej

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LEAVE ANY  
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

## 4. Galleria\_Mall

Production • 61.3s

**Question:** Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

**Error reason:** AI indicated failure (matched 'unexpected keyword argument'): Clipping Loss Analysis (2026-06-07 to 2026-08-07) - Requested breakdown: per inverter (Inverter 1 through Inverter 12) - Data quality note: clipping-loss calculation tool failed internally •

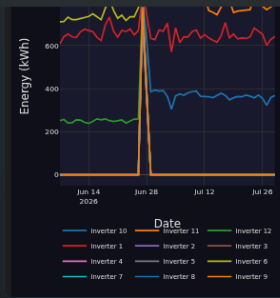
## TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

### Raw API response:

[illegible]

**Console errors:** An iframe which has both allow-scripts and allow-same-origin for its sandbox attribute can escape its sandboxing.



[View Fullscreen](#)

01:24 AM

Are you satisfied with the answer?



Time taken: 45s, token: 0 tokens

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LEAVE ANY  
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

## 5. GID

Production • 114.4s

**Question:** Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

**Error reason:** AI indicated failure (matched 'internal error'): Data Calculation Results: - Clipping-loss calculation: Failed due to an internal error. • Error message: `calculate\_clipping\_or\_curtailment\_losses() got an unexpected keyword argument 'reactive\_col`

## TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well",
    "displayed_response": "Data Calculation Results:\n\n- Clipping-loss calculation: Failed due to an internal error.\n  · Error message: `calculate_clipping_or_curtailment_losses()` got an unexpected keyword argument 'reactive_col'\n- Reliable clipping-loss values per inverter for the last 6 months are unavailable.\n\nVisualization:\n\n- The visualization for “Clipping Loss by Inverter – Last 6 Months” has been prepared.",
    "coding_response": [
      "I couldn't calculate clipping loss per inverter because the clipping-loss tool failed internally.\n\nThe tool error was:\n`calculate_clipping_or_curtailment_losses()` got an unexpected keyword argument 'reactive_col'\n\nSo I'm not able to provide reliable clipping-loss values from the available tooling right now.",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping Loss by Inverter – Last 6 Months",
        "yAxisTitle": "Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "functions.calculate_clipping_or_curtailment_losses_ag",
      "functions.total_energy_func_ag",
      "functions.inspect_xpool"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from February 7, 2026, to August 7, 2026, including the last day.\n2. I find all columns that represent inverter power, making sure to leave out any columns related to reactive power or flags.\n3. I clean each inverter's power data by removing or marking any invalid measurements.\n4. For each inverter, I calculate the energy produced by multiplying the power values by the time interval (0.0833 hours), and convert units if needed (for example, dividing by 1000 if the data is in Watts).\n5. I make sure to include every day in the calculation for each inverter.\n6. I sum up the energy values for each inverter, grouping them by the chosen time period (such as daily, monthly, or yearly).\n7. I provide the energy totals for each time period, broken down by inverter.\n8. I organize the results into a DataFrame, with each inverter as a column and each time period as a row.\n9. I return the results as a dictionary, using the time period as the key, or as a single total row, depending on the settings.\n\n1. I filter the data to include only records from February 7, 2026, to August 7, 2026, including the last day.\n2. I identify all valid inverter power columns, excluding those related to reactive power or flags.\n3. I check that the specified power columns are present in the dataset.\n4. I clean the plane of array (POA) irradiance data by filling in missing values using a diurnal pattern to maintain data quality.\n5. For each inverter, I calculate power losses as the absolute difference between the measured power and the inverter's rated AC power during periods of detected loss.\n  - I detect clipping events using a combination of machine learning (rdtools XGBoost clip filter) and threshold logic, flagging power values above 85% of the rated AC power.\n6. I convert the detected power losses to energy lost (in kWh) by multiplying by the time interval (0.0833 hours).\n7. I sum up the energy losses for each inverter, grouping them by the chosen time period (such as daily, monthly, or yearly).\n8. I provide the loss totals for each time period, broken down by inverter.\n9. I organize the results into a DataFrame, with each inverter as a column and each time period as a row.\n10. I return the results as a dictionary, using the time period as the key, or as a single total row, depending on the settings."
  }
}
```



### Data Calculation Results:

- Clipping-loss calculation: Failed due to an internal error.
- Error message:  
`calculate\_clipping\_or\_curtailment\_losses()` got an unexpected keyword argument 'reactive\_col'
- Reliable clipping-loss values per inverter for the last 6 months are unavailable.

### Visualization:

- The visualization for "Clipping Loss by Inverter — Last 6 Months" has been prepared.

01:36 AM



Are you satisfied with the answer?



Time taken: 99s, token: 0 tokens

connected to Mathkor Gid



How can we help you today?



LEAVE ANY  
FEEDBACK



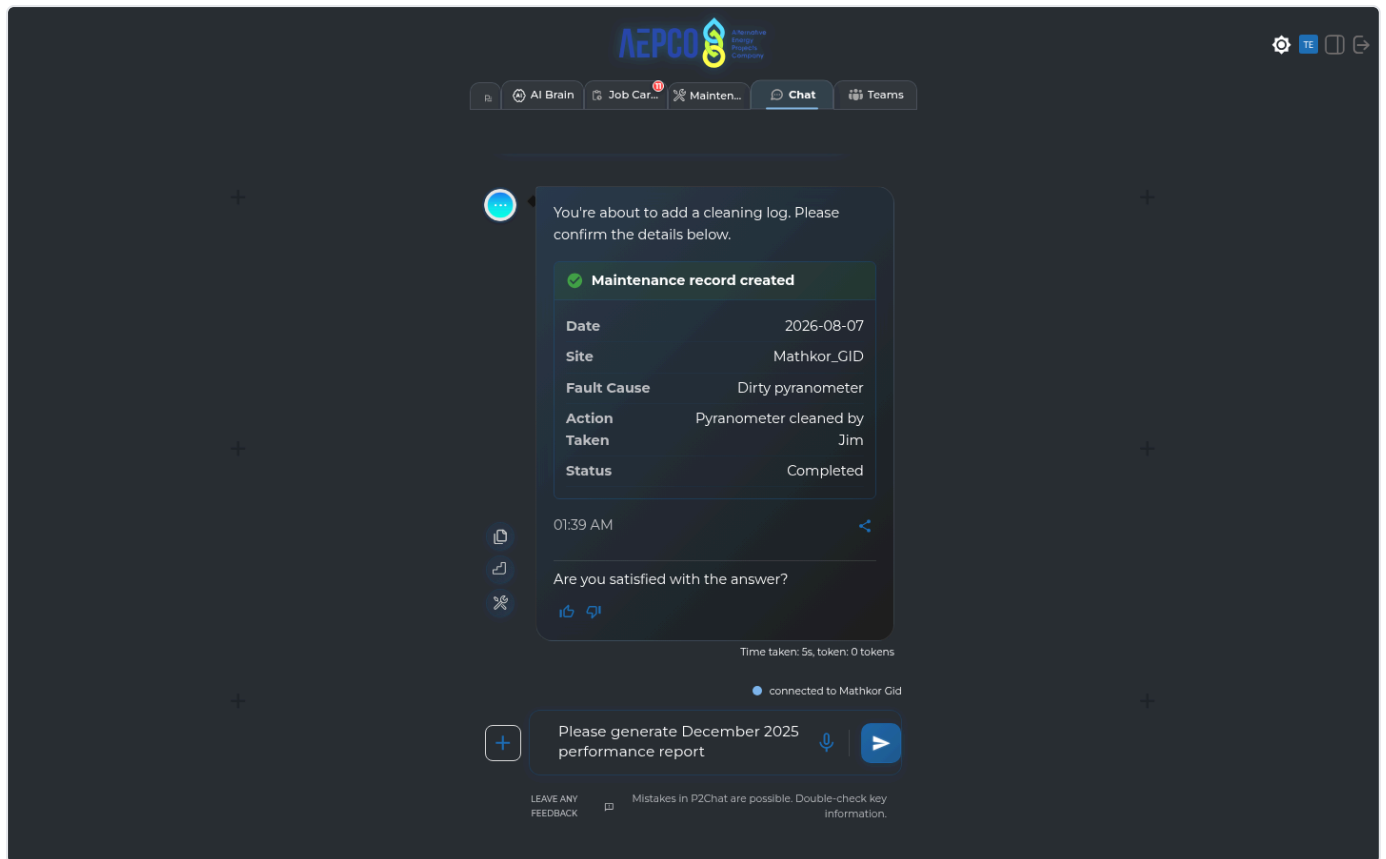
Mistakes in P2Chat are possible. Double-check key information.

## 6. GID

Production • 17.0s

**Question:** Please generate December 2025 performance report

**Error reason:** AI indicated failure (matched 'internal error'): AI Brain Job Cards 11 Maintenance Chat Teams Active Power is 0W Wind Speed is 1.3m/s Weather Temperature is 32°C Last updated: 8/7/2026, 12:00 AM 01:28 AM Good evening. How can I help yo



## 7. Abdali\_Mall\_Netmetering

Production • 55.1s

**Question:** Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

**Error reason:** AI indicated failure (matched 'internal error'): Below is the summary for clipping losses over the last 6 months, broken down by inverter: Data Calculation: - Unable to compute per-inverter clipping losses due to an internal error in the clipping-I



## TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well",
    "displayed_response": "Below is the summary for clipping losses over the last 6 months, broken down by inverter:\n\nData Calculation:\n- Unable to compute per-inverter clipping losses due to an internal error in the clipping-loss tool ('reactive_col' parameter error).\n- No reliable clipping loss values are available for the requested period and inverters.\n\nVisualization:\n- The graph illustrating clipping losses by inverter for the last 6 months has been prepared.",
    "coding_response": [
      "I couldn't calculate the clipping losses because the clipping-loss tool is currently failing internally for this plant/time range with an unexpected parameter error ('reactive_col'). I'm therefore unable to return reliable per-inverter clipping loss values for the last 6 months.",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping Loss by Inverter for the Last 6 Months",
        "yAxisTitle": "Clipping Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "functions.inspect_xpool",
      "functions.calculate_clipping_or_curtailment_losses_ag"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from February 7, 2026, to August 7, 2026, including the end date.\n2. I find all relevant inverter power columns in the dataset, making sure to leave out any columns related to reactive power or flags.\n3. I check that the chosen power columns are actually present in the dataset.\n4. I clean up the plane of array (POA) irradiance data by filling in any missing values using a method that accounts for daily sunlight patterns, to keep the data accurate.\n5. For each inverter power column, I calculate power losses by finding the absolute difference between the measured power and the inverter's rated AC power during times when losses are detected.\n6. I identify clipping events using both a machine learning approach (rdtools XGBoost clip filter) and a simple rule: if the power is above 85% of the rated AC power, it's considered a potential clipping event.\n7. I convert the detected power losses into energy lost (in kWh) by multiplying the loss by the time interval (0.0833 hours).\n8. I group the energy loss values by the chosen time period (such as day, month, or year) for each inverter.\n9. I provide a detailed breakdown of the loss values for each inverter and each time period.\n10. I organize the results into a DataFrame, with each inverter as a column and each time period as a row.\nI return the results as a dictionary, either broken down by time period or as a single total, depending on the settings."
  }
}
```



Below is the summary for clipping losses over the last 6 months, broken down by inverter:

Data Calculation:

- Unable to compute per-inverter clipping losses due to an internal error in the clipping-loss tool ("reactive\_col" parameter error).
- No reliable clipping loss values are available for the requested period and inverters.

Visualization:

- The graph illustrating clipping losses by inverter for the last 6 months has been prepared.

01:54 AM



Are you satisfied with the answer?



Time taken: 39s, token: 0 tokens

connected to Abdali Mall Netmetering



How can we help you today?



LEAVE ANY  
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

## 8. Abdali\_Mall\_Netmetering

Production • 56.7s

**Question:** Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

**Error reason:** AI indicated failure (matched 'tool error'): Data Calculation & Quality Notes: - Unable to compute per-inverter clipping loss due to an internal tool failure. - Plant AC power (for reference): 650 kW. - Metadata vs. data inconsistency:

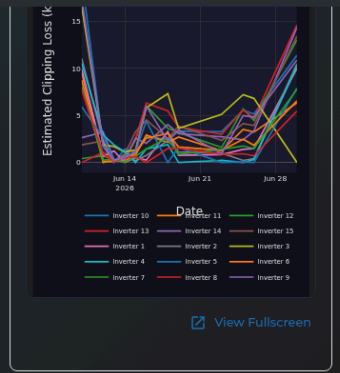
## TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

### Raw API response:

[illegible]

**Console errors:** An iframe which has both allow-scripts and allow-same-origin for its sandbox attribute can escape its sandboxing.



02:02 AM

Are you satisfied with the answer?



Time taken: 41s, token: 0 tokens

connected to Abdali Mall Netmetering



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LEAVE ANY  
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

## 9. BF\_Wheeling

Production • 52.0s

**Question:** Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

**Error reason:** AI indicated failure (matched 'tool error'): Clipping Loss Analysis – Last 6 Months (Inverter Breakdown) • Data Calculation – No clipping loss values could be calculated for the last 6 months. – Reason: Internal clipping-loss tool failure w

## TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well",
    "displayed_response": "Clipping Loss Analysis - Last 6 Months (Inverter Breakdown)\n\n· Data Calculation\n- No clipping loss values could be calculated for the last 6 months.\n- Reason: Internal clipping-loss tool failure with an unexpected-argument error.\n- Data Quality Note: No reliable inverter-specific clipping-loss data available due to the tool error.\n\n· Visualization\n- The chart “Clipping Loss by Inverter - Last 6 Months” has been prepared. Note that it does not display actual values because the underlying data could not be generated.",
    "coding_response": [
      "I couldn't calculate clipping loss because the clipping-loss tool is currently failing internally for this plant/time range with an unexpected-argument error. No clipping-loss values were returned, so I can't provide a reliable inverter breakdown.",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping Loss by Inverter - Last 6 Months",
        "yAxisTitle": "Clipping Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "functions.calculate_clipping_or_curtailment_losses_ag",
      "functions.inspect_xpool"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from February 7, 2026, to August 7, 2026, including the last day.\n2. I identify all relevant inverter power columns in the dataset, making sure to leave out any columns related to reactive power or flags.\n3. I check that the specified power columns are actually present in the data.\n4. I clean up the plane of array (POA) irradiance data by filling in any missing values using a method that accounts for daily sunlight patterns.\n5. For each inverter power column, I calculate power losses by finding the absolute difference between the measured power and the inverter's rated AC power during periods of detected loss.\n- I detect clipping events using both a machine learning approach (rdtools XGBoost clip filter) and a rule that flags power values above 85% of the inverter's rated AC power.\n6. I convert the detected power losses into energy lost (in kWh) by multiplying by the time interval of 0.0833 hours.\n7. I sum up the energy losses for each inverter, grouping the results by month (or by day, month, or year as needed).\n8. I provide a detailed breakdown of the loss values for each inverter and each time period.\n9. I organize the results into a DataFrame, with each inverter as a column and each time period as a row.\n10. I return the results as a dictionary, where each key is a time period (or a single total row), depending on the chosen settings."
  }
}
```

Clipping Loss Analysis – Last 6 Months  
(Inverter Breakdown)

- Data Calculation
  - No clipping loss values could be calculated for the last 6 months.
  - Reason: Internal clipping-loss tool failure with an unexpected-argument error.
  - Data Quality Note: No reliable inverter-specific clipping-loss data available due to the tool error.

- Visualization
  - The chart "Clipping Loss by Inverter – Last 6 Months" has been prepared. Note that it does not display actual values because the underlying data could not be generated.

02:12 AM

Are you satisfied with the answer?



Time taken: 36s, token: 0 tokens

connected to BF Wheeling

+ How can we help you today?

LEAVE ANY  
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

## 10. Abdali\_Mall\_Barza

Production • 47.5s

**Question:** Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

**Error reason:** AI indicated failure (matched 'unexpected keyword argument'): Data Retrieval Issue - Period analyzed: 2026-06-07 to 2026-08-07 - Breakdown requested: by inverter - Outcome: clipping-loss tool failed internally and could not produce results - Error reported: une

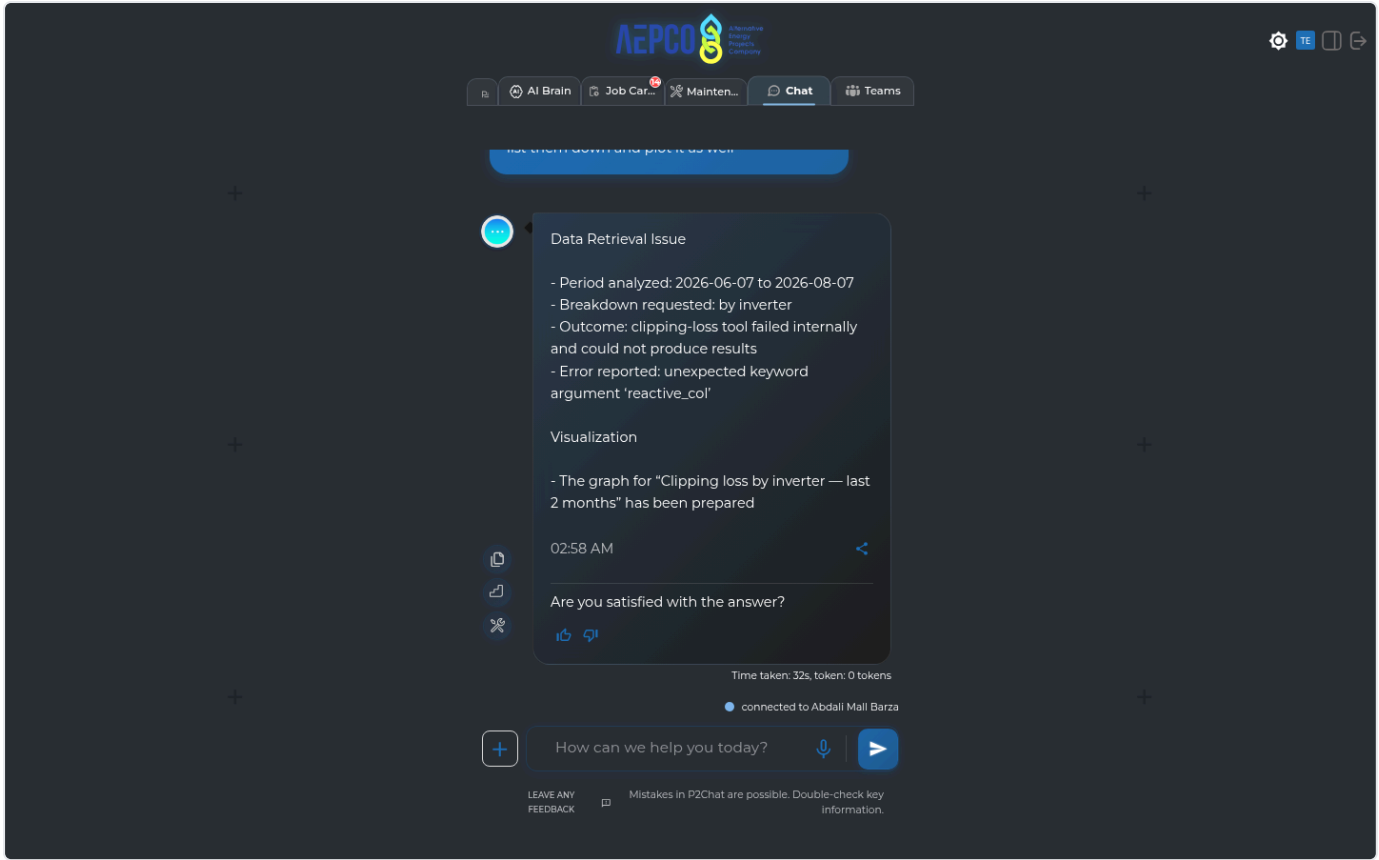


## TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well",
    "displayed_response": "Data Retrieval Issue\n\n- Period analyzed: 2026-06-07 to 2026-08-07\n- Breakdown requested: by inverter\n- Outcome: clipping-loss tool failed internally and could not produce results\n- Error reported: unexpected keyword argument 'reactive_col'\n\nVisualization\n\n- The graph for \"Clipping loss by inverter – last 2 months\" has been prepared",
    "coding_response": [
      "I couldn't calculate clipping loss because the clipping-loss tool is currently failing internally for this plant/query. It returns an exception before producing any inverter results.\n\nAttempted period: 2026-06-07 to 2026-08-07\nRequested breakdown: by inverter\nIssue returned by tool: unexpected keyword argument 'reactive_col'",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping loss by inverter – last 2 months",
        "yAxisTitle": "Clipping Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "functions.inspect_xpool",
      "functions.calculate_clipping_or_curtailment_losses_ag",
      "functions.total_energy_func_ag"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from June 7, 2026, to August 7, 2026, including the end date.\n2. I find all relevant inverter power columns in the dataset, making sure to leave out any columns related to reactive power or flags.\n3. I check that the required power columns are actually present in the data.\n4. I clean up the plane of array (POA) irradiance data by filling in any missing values using a method that accounts for daily sunlight patterns.\n5. For each inverter, I calculate power losses by taking the absolute difference between the measured power and the inverter's rated AC power during periods when losses are detected. I identify these loss periods (clipping events) using a mix of machine learning (rdtools XGBoost clip filter) and a rule that flags power above 85% of the rated AC power.\n6. I convert these power losses into energy lost (in kWh) by multiplying the loss values by the time interval (0.0833 hours).\n7. I group the energy loss values by the chosen time period (such as day, month, or year) for each inverter.\n8. I provide a detailed breakdown of the loss values for each inverter and each time period.\n9. I organize the results into a DataFrame, with each inverter as a column and each time period as a row.\n10. I return the results as a dictionary, where each key is a time period (or a single total row), depending on the settings."
  }
}
```



# LLM Reasoning Issues LLM REASONING (9 failures)

**Question:** When was the last time the BOM data showed up

**Error reason:** AI indicated failure (matched 'i'm sorry'): I'm sorry, I can't help with plant performance or data report queries. Please use the report features or ask for a plant report if you need BOM data history or details. 10:30 AM Are you satisfied wi

#### TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "When was the last time the BOM data showed up",
    "displayed_response": "I'm sorry, I can't help with plant performance or data report queries. Please use the report features or ask for a plant report if you need BOM data history or details.",
    "coding_response": [
      [
        {
          "answer": "I'm sorry, I can't help with plant performance or data report queries. Please use the report features or ask for a plant report if you need BOM data history or details.",
          "action": []
        }
      ]
    ],
    "used_agents": [
      "ui_agent"
    ],
    "used_tools": [],
    "urls": [],
    "steps": []
  }
}
```



Are you satisfied with the answer?

Time taken: 6s, token: 0 tokens

When was the last time the BOM data showed up



I'm sorry, I can't help with plant performance or data report queries. Please use the report features or ask for a plant report if you need BOM data history or details.

10:30 AM

Are you satisfied with the answer?

Time taken: 9s, token: 0 tokens



How can we help you today?

**Question:** How can I shut down my inverters?

**Error reason:** AI indicated failure (matched 'unable to provide'): The manual documents for your system are not available or not mapped, so I am unable to provide shutdown instructions for your inverters from the manuals. If you have another source or more details, p

#### TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "How can I shut down my inverters?",
    "displayed_response": "The manual documents for your system are not available or not mapped, so I am unable to provide shutdown instructions for your inverters from the manuals. If you have another source or more details, please provide them, or please contact your service provider for assistance.",
    "coding_response": [
      "The manual documents for your system are not available or not mapped, so I am unable to provide shutdown instructions for your inverters from the manuals. If you have another source or more details, please provide them, or please contact your service provider for assistance."
    ],
    "used_agents": [
      "manual_agent"
    ],
    "used_tools": [
      "functions.manual_rag_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```

Are you satisfied with the answer?

👍 🗨️

Time taken: 9s, token: 0 tokens

How can I shut down my inverters? **TE**

...

The manual documents for your system are not available or not mapped, so I am unable to provide shutdown instructions for your inverters from the manuals. If you have another source or more details, please provide them, or please contact your service provider for assistance.

10:47 AM

Are you satisfied with the answer?

👍 🗨️

Time taken: 5s, token: 0 tokens

● connected to Mof

+ How can we help you today? 🗣️ ➡️

**Question:** Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

**Error reason:** AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 01:04 AM Do you want to report this error? YES NO [Click here for Source](#) [Click here for Steps](#)

TECHNICAL EVIDENCE

|                   |                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------|
| Exception         | PydanticSerializationError: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'> |
| Agent/Tool        | data_agent / unknown                                                                                |
| Endpoint          | /testuser1/ai_agent_chat                                                                            |
| Location          | agent_graph.py:916                                                                                  |
| API call:         | GET /testuser2/ai_agent_chat → 400                                                                  |
| Raw API response: |                                                                                                     |

```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.",
  "coding_response": [
    "===== ERROR SUMMARY =====\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGENT: data_agent\nTOOL: unknown\nERROR TYPE: PydanticSerializationError\nERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>\nLOCATION: agent_graph.py:916\n======"
  ]
}
```

```
===== ERROR SUMMARY =====
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: data_agent
TOOL: unknown
ERROR TYPE: PydanticSerializationError
ERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>
LOCATION: agent_graph.py:916
=====
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)

111112 111111 1111 111111 111111 111111 111111 111111 111111 111111

Is there shading happening with my plant?  
Please analyze the pyranometer and check for frequent dips at certain times

TE



Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.

01:04 AM



Do you want to report this error?

YES NO



< Click here for Source



< Click here for Steps



< Click here for Tools

Time taken: s, token: tokens

connected to Kupco Hq Dajeej



How can we help you today?



LEAVE ANY  
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.



## 4. GID

**Question:** Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3

**Error reason:** AI indicated failure (matched 'an error'): Sorry, there was an error processing your request.

01:31 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected to Mathkor Gid  
LEAVE ANY FEEDBACK Mistakes in P2Chat

## TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{ "result": { "user_input": "\nPlease plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3\n",\n    \"displayed_response\": \"I've created the visualization for you.\",\n    \"coding_response\": [\n        {\n            \"type\": \"graph\",\n            \"series\": [\n                <html><head><meta charset=\\\"utf-8\\\" /></head><nbody>\n                    <div>\n                        <script type=\\\"text/javascript\\\"\n                            >window.PlotlyConfig = {MathJaxConfig: 'local'};</script>\n                        <script charset=\\\"utf-8\\\" src=\\\"https://cdn.plot.ly/plotly-3.0.1.min.js\\\"\n                            ></script>\n                        <div id=\\\"e26055dc-8fcd-4fa1-a88e-ca69bd3e3f91\\\" class=\\\"plotly-graph-div\\\" style=\\\"height:100%; width:100%;\\\">\n                            <script type=\\\"text/javascript\\\"\n                                window.PLOTLYENV=window.PLOTLYENV || {};\n                                if (document.getElementById(\\\"e26055dc-8fcd-4fa1-a88e-ca69bd3e3f91\\\")) {\n                                    Plotly.newPlot(\n                                        \\\"e26055dc-8fcd-4fa1-a88e-ca69bd3e3f91\\\", [\n                                            [\\\"hovertemplate\\\": \\\"%{fullData.name}: %y:.2f\\\" A\\\"cextra\\\"u003e\\\"u003c\\\"u002ffextra\\\"u003e\\\", \\\"mode\\\": \\\"lines\\\", \\\"name\\\": \\\"Inverter 3 mpt10 Current\\\", \\\"x\\\": [\n                                                \"2026-08-06T00:00:00\", \"2026-08-06T00:05:00\", \"2026-08-06T00:10:00\", \"2026-08-06T00:15:00\", \"2026-08-06T00:20:00\", \"2026-08-06T00:25:00\", \"2026-08-06T00:30:00\", \"2026-08-06T00:35:00\", \"2026-08-06T00:40:00\", \"2026-08-06T00:45:00\", \"2026-08-06T00:50:00\", \"2026-08-06T00:55:00\", \"2026-08-06T01:00:00\", \"2026-08-06T01:05:00\", \"2026-08-06T01:10:00\", \"2026-08-06T01:15:00\", \"2026-08-06T01:20:00\", \"2026-08-06T01:25:00\", \"2026-08-06T01:30:00\", \"2026-08-06T01:35:00\", \"2026-08-06T01:40:00\", \"2026-08-06T01:45:00\", \"2026-08-06T01:50:00\", \"2026-08-06T01:55:00\", \"2026-08-06T02:00:00\", \"2026-08-06T02:05:00\", \"2026-08-06T02:10:00\", \"2026-08-06T02:15:00\", \"2026-08-06T02:20:00\", \"2026-08-06T02:25:00\", \"2026-08-06T02:30:00\", \"2026-08-06T02:35:00\", \"2026-08-06T02:40:00\", \"2026-08-06T02:45:00\", \"2026-08-06T02:50:00\", \"2026-08-06T02:55:00\", \"2026-08-06T03:00:00\", \"2026-08-06T03:05:00\", \"2026-08-06T03:10:00\", \"2026-08-06T03:15:00\", \"2026-08-06T03:20:00\", \"2026-08-06T03:25:00\", \"2026-08-06T03:30:00\", \"2026-08-06T03:35:00\", \"2026-08-06T03:40:00\", \"2026-08-06T03:45:00\", \"2026-08-06T03:50:00\", \"2026-08-06T03:55:00\", \"2026-08-06T04:00:00\", \"2026-08-06T04:05:00\", \"2026-08-06T04:10:00\", \"2026-08-06T04:15:00\", \"2026-08-06T04:20:00\", \"2026-08-06T04:25:00\", \"2026-08-06T04:30:00\", \"2026-08-06T04:35:00\", \"2026-08-06T04:40:00\", \"2026-08-06T04:45:00\", \"2026-08-06T04:50:00\", \"2026-08-06T04:55:00\", \"2026-08-06T05:00:00\", \"2026-08-06T05:05:00\", \"2026-08-06T05:10:00\", \"2026-08-06T05:15:00\", \"2026-08-06T05:20:00\", \"2026-08-06T05:25:00\", \"2026-08-06T05:30:00\", \"2026-08-06T05:35:00\", \"2026-08-06T05:40:00\", \"2026-08-06T05:45:00\", \"2026-08-06T05:50:00\", \"2026-08-06T05:55:00\", \"2026-08-06T06:00:00\", \"2026-08-06T06:05:00\", \"2026-08-06T06:10:00\", \"2026-08-06T06:15:00\", \"2026-08-06T06:20:00\", \"2026-08-06T06:25:00\", \"2026-08-06T06:30:00\", \"2026-08-06T06:35:00\", \"2026-08-06T06:40:00\", \"2026-08-06T06:45:00\", \"2026-08-06T06:50:00\", \"2026-08-06T07:00:00\", \"2026-08-06T07:05:00\", \"2026-08-06T07:10:00\", \"2026-08-06T07:15:00\", \"2026-08-06T07:20:00\", \"2026-08-06T07:25:00\", \"2026-08-06T07:30:00\", \"2026-08-06T07:35:00\", \"2026-08-06T07:40:00\", \"2026-08-06T07:45:00\", \"2026-08-06T07:50:00\", \"2026-08-06T07:\n                                                    ]\n                                        ]\n                                    }\n                                }\n                            </script>\n                        \n                    \n                \n            ]\n        }]\n    ]\n}
```

**Console errors:** An iframe which has both allow-scripts and allow-same-origin for its sandbox attribute can escape its sandboxing. | Access to fetch at 'https://weortmh7ksmbdtyoqh5wjhea440blqr. lambda-url.ap-southeast-2.on.aws/testuser2/update\_plant\_history' from origin 'https://aepc2.p2agentx.com' has been blocked

**Network failures:** POST https://weortmh7ksmbdtyoqh5wjhea440blqru.lambda-url.ap-southeast-2.on.aws/testuser2/update\_plant\_history (net::ERR\_FAILED)



- < Click here for Source
- < Click here for Steps
- < Click here for Tools

Time taken: 15s, token: 0 tokens

Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3



Sorry, there was an error processing your request.

01:31 AM



Do you want to report this error?

YES NO

Time taken: s, token: tokens

connected to Mathkor Gid



How can we help you today?



LEAVE ANY  
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

**Question:** Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

**Error reason:** AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 01:42 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected t

TECHNICAL EVIDENCE

|            |                                                                |
|------------|----------------------------------------------------------------|
| Exception  | AttributeError: 'NoneType' object has no attribute 'rule_code' |
| Agent/Tool | data_agent / extract_timeseries_data_ag                        |
| Endpoint   | /testuser1/ai_agent_chat                                       |
| Location   | agent_graph.py:916                                             |

API call: GET /testuser2/ai\_agent\_chat → 400

Raw API response:

```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified
and will investigate.",
  "coding_response": [
    "===== ERROR SUMMARY =====\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGEN
T: data_agent\nTOOL: extract_timeseries_data_ag\nERROR TYPE: AttributeError\nERROR MESSAGE: 'NoneType' obje
ct has no attribute 'rule_code'\nLOCATION: agent_graph.py:916\n=====
  ]
}
```

```
===== ERROR SUMMARY =====
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: data_agent
TOOL: extract_timeseries_data_ag
ERROR TYPE: AttributeError
ERROR MESSAGE: 'NoneType' object has no attribute 'rule_code'
LOCATION: agent_graph.py:916
=====
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)



Are you satisfied with the answer?



Time taken: 39s, token: 0 tokens

Is there shading happening with my plant?  
Please analyze the pyranometer and check for frequent dips at certain times



Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.

01:42 AM



Do you want to report this error?

YES NO

Time taken: s, token: tokens

● connected to Mathkor Gid



How can we help you today?



LEAVE ANY  
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

**Question:** Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

**Error reason:** AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 01:43 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected t

TECHNICAL EVIDENCE

|            |                                                                |
|------------|----------------------------------------------------------------|
| Exception  | AttributeError: 'NoneType' object has no attribute 'rule_code' |
| Agent/Tool | data_agent / extract_timeseries_data_ag                        |
| Endpoint   | /testuser1/ai_agent_chat                                       |
| Location   | agent_graph.py:916                                             |

API call: GET /testuser2/ai\_agent\_chat → 400

Raw API response:

```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.",
  "coding_response": [
    "===== ERROR SUMMARY =====\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGENT: data_agent\nTOOL: extract_timeseries_data_ag\nERROR TYPE: AttributeError\nERROR MESSAGE: 'NoneType' object has no attribute 'rule_code'\nLOCATION: agent_graph.py:916\n=====
  ]
}
```

```
===== ERROR SUMMARY =====
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: data_agent
TOOL: extract_timeseries_data_ag
ERROR TYPE: AttributeError
ERROR MESSAGE: 'NoneType' object has no attribute 'rule_code'
LOCATION: agent_graph.py:916
=====
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)



Do you want to report this error?

YES NO

Time taken: s, token: tokens

Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well



Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.

01:43 AM



Do you want to report this error?

YES NO

Time taken: s, token: tokens

● connected to Mathkor Gid



How can we help you today?



LEAVE ANY  
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

**Question:** Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

**Error reason:** AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 02:01 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected t

TECHNICAL EVIDENCE

Exception    PydanticSerializationError: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>

Agent/Tool   data\_agent / unknown

Endpoint    /testuser1/ai\_agent\_chat

Location    agent\_graph.py:916

API call: GET /testuser2/ai\_agent\_chat → 400

Raw API response:

```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.",
  "coding_response": [
    "===== ERROR SUMMARY =====\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGENT: data_agent\nTOOL: unknown\nERROR TYPE: PydanticSerializationError\nERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>\nLOCATION: agent_graph.py:916\n=====
  ]
}
```

```
===== ERROR SUMMARY =====
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: data_agent
TOOL: unknown
ERROR TYPE: PydanticSerializationError
ERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>
LOCATION: agent_graph.py:916
=====
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)



Are you satisfied with the answer?



Time taken: 27s, token: 0 tokens

Is there shading happening with my plant?  
Please analyze the pyranometer and check for frequent dips at certain times



Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.

02:01 AM



Do you want to report this error?

YES NO

Time taken: s, token: tokens

connected to Abdali Mall Netmetering



How can we help you today?



LEAVE ANY  
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.



**Question:** I have a COMMUNICATION\_FAULT what do I do to my inverters?

**Error reason:** AI indicated failure (matched 'unable to provide'): The manual documents do not contain relevant information regarding the "COMMUNICATION\_FAULT" issue for your inverters. As a result, I am unable to provide troubleshooting steps or diagram references f

#### TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "I have a COMMUNICATION_FAULT what do I do to my inverters?",
    "displayed_response": "The manual documents do not contain relevant information regarding the \"COMMUNICATION_FAULT\" issue for your inverters. As a result, I am unable to provide troubleshooting steps or diagram references from the manuals.",
    "coding_response": [
      "The manual documents do not contain relevant information regarding the \"COMMUNICATION_FAULT\" issue for your inverters. As a result, I am unable to provide troubleshooting steps or diagram references from the manuals."
    ],
    "used_agents": [
      "manual_agent"
    ],
    "used_tools": [
      "functions.manual_rag_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```

Are you satisfied with the answer?

Time taken: 6s, token: 0 tokens

I have a COMMUNICATION\_FAULT what do I do to my inverters?

TE

The manual documents do not contain relevant information regarding the "COMMUNICATION\_FAULT" issue for your inverters. As a result, I am unable to provide troubleshooting steps or diagram references from the manuals.

02:48 AM

Are you satisfied with the answer?

Time taken: 5s, token: 0 tokens

connected to Abdali Mali Barza

How can we help you today?

## 9. Abdali\_Mall\_Barza

Production • 41.3s

**Question:** Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

**Error reason:** AI indicated failure (matched 'cannot provide'): Data Calculation: - The clipping-loss calculation for the last 6 months could not be completed due to an internal exception in the clipping-loss tool. - No validated clipping-loss values were returned

## TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well",
    "displayed_response": "Data Calculation:\n\n- The clipping-loss calculation for the last 6 months could not be completed due to an internal exception in the clipping-loss tool.\n- No validated clipping-loss values were returned.\n- As a result, we cannot provide an inverter-by-inverter breakdown for the requested period.\n\nVisualization:\n\n- The visualization titled \"Clipping Loss per Inverter - Last 6 Months\" has been prepared.\n- Since no data was available, the graph contains no series values.",
    "coding_response": [
      "I couldn't complete the clipping-loss calculation because the clipping-loss tool failed internally with an exception. No validated clipping-loss values were returned, so I can't provide a reliable inverter-by-inverter breakdown for the last 6 months.",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping Loss per Inverter - Last 6 Months",
        "yAxisTitle": "Clipping Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "calculate_clipping_or_curtailement_losses_ag",
      "total_energy_func_ag",
      "functions.inspect_xpool",
      "functions.calculate_clipping_or_curtailement_losses_ag"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from February 7, 2026, to August 7, 2026, including the end date.\n2. I identify all relevant inverter power columns in the dataset, making sure to exclude any columns related to reactive power or flags.\n3. I check that the required inverter power columns are present in the data.\n4. I clean the plane of array (POA) irradiance data by filling in missing values using a method that accounts for daily sunlight patterns.\n5. For each inverter, I calculate energy losses by finding the absolute difference between the measured power and the inverter's rated AC power during periods of detected losses. \n- I detect clipping events using both a machine learning model (rdtools XGBoost clip filter) and a rule that flags power readings above 85% of the inverter's rated AC power.\n6. I convert the detected power losses into energy lost (in kWh) by multiplying by the time interval of 0.0833 hours.\n7. I group the loss energy values by month (or by day, month, or year, depending on the chosen aggregation) for each inverter.\n8. I provide a detailed breakdown of the loss values for each inverter and each aggregation period.\n9. I organize the results into a DataFrame, with each inverter as a column and each aggregation period as a row.\n10. I return the results as a dictionary, either broken down by aggregation period or as a single total row, depending on the settings."
  }
}
```



## 1. BF\_Wheeling

Production • 108.1s

**Question:** Please generate last months report

**Error reason:** AI indicated failure (matched 'error occurred'): Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF\_Wheeling' was not found in external\_report\_sections.csv. Available plants: KHBP, SK\_EX, SK\_Wheeling, HPP, Noor\_AI

### TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please generate last months report",
    "displayed_response": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AL_Mushagar\"",
    "coding_response": [
      [
        {
          "answer": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AL_Mushagar\"",
          "action": []
        }
      ]
    ],
    "used_agents": [
      "report_agent"
    ],
    "used_tools": [
      "agent_external_report_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```



Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF\_Wheeling' was not found in external\_report\_sections.csv. Available plants: KHBP, SK\_EX, SK\_Wheeling, HPP, Noor\_AL\_Aqaba, Abdali\_Mall\_Netmetering, Mathkor\_GID, MOF, Galleria\_Mall, KUPCO\_HQ\_Dajeej, SADAFCO, Abdali\_Mall\_Lahoun, Abdali\_Mall\_Barza, Abdali\_Mall\_AI\_Mushagar'

02:05 AM



Are you satisfied with the answer?



- < Click here for Source
- < Click here for Steps
- < Click here for Tools

Time taken: 93s, token: 0 tokens

● connected to BF Wheeling



How can we help you today?



LEAVE ANY  
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

## 2. BF\_Wheeling

Production • 46.0s

**Question:** Please generate December 2025 performance report

**Error reason:** AI indicated failure (matched 'error occurred'): Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF\_Wheeling' was not found in external\_report\_sections.csv.

Available plants: KHBP, SK\_EX, SK\_Wheeling, HPP, Noor\_AI

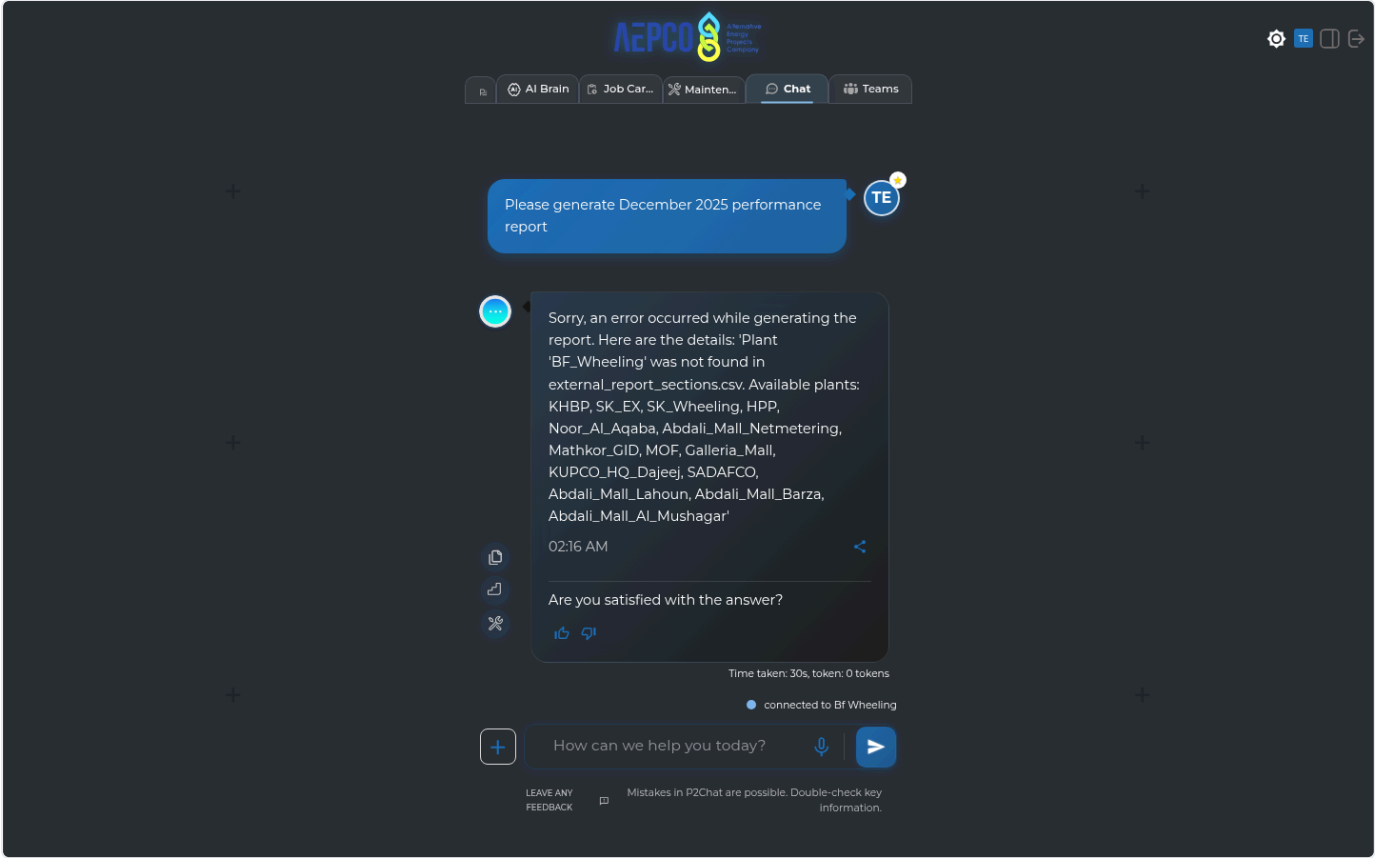
### TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please generate December 2025 performance report",
    "displayed_response": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AL_Mushagar'",
    "coding_response": [
      [
        {
          "answer": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AL_Mushagar'",
          "action": []
        }
      ]
    ],
    "used_agents": [
      "report_agent"
    ],
    "used_tools": [
      "agent_external_report_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```





**Graph/Visualization Errors** GRAPH VISUALIZATION (1 failure)

**Question:** Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

**Error reason:** AI indicated failure (matched 'failed to'): Data Quality Note: - The clipping-loss tool encountered an internal error (`unexpected keyword argument 'reactive\_col'`) and failed to produce per-inverter values for the last 2 months. Visualiz

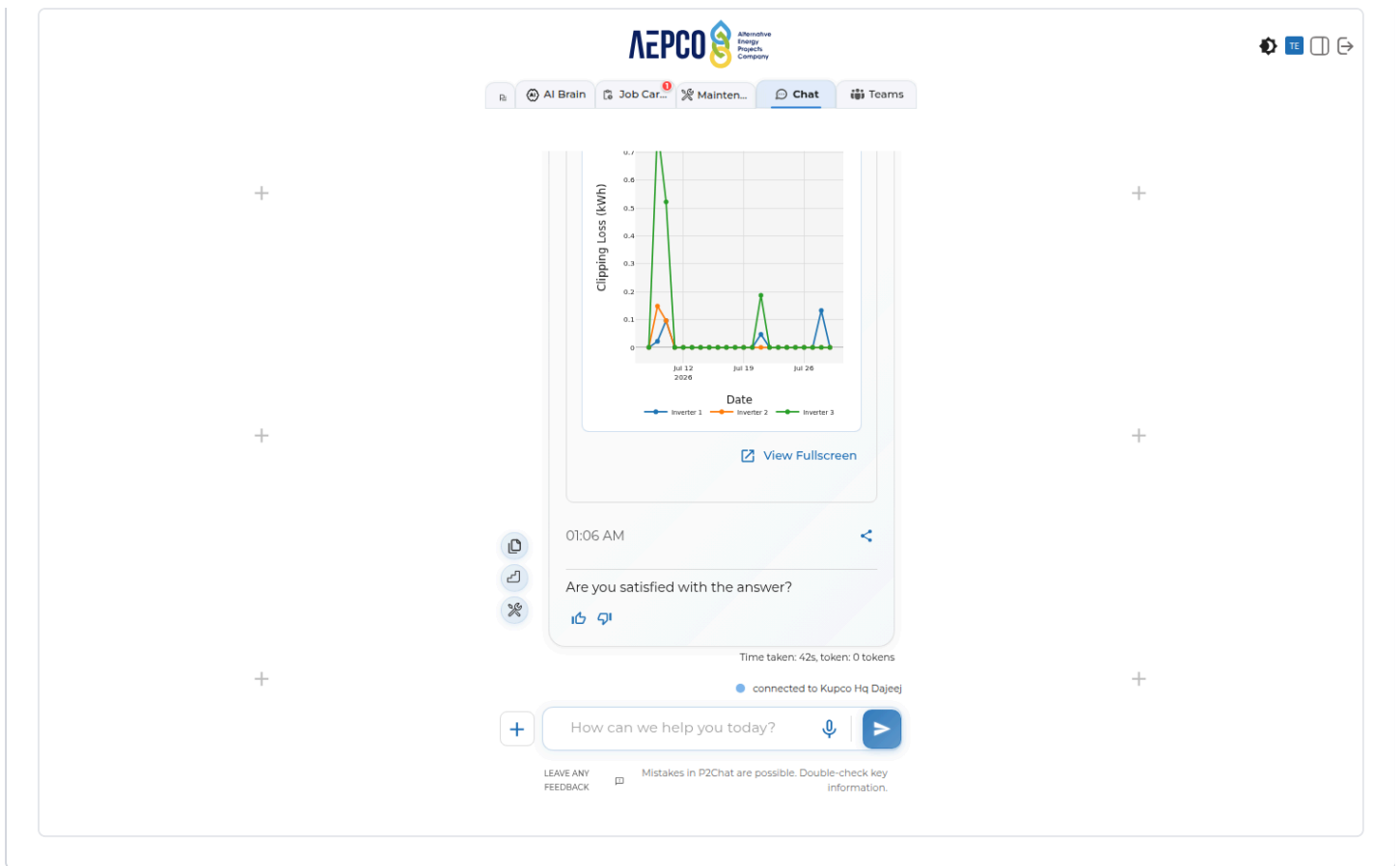
## TECHNICAL EVIDENCE

API call: GET /testuser2/ai\_agent\_chat → 200

### Raw API response:

[illegible]

**Console errors:** An iframe which has both allow-scripts and allow-same-origin for its sandbox attribute can escape its sandboxing.



## Skipped (21)

| Plant | Question                                                                                                                                                                      | Env        | Skip reason             |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------|
| SAK_1 | Please generate last months report                                                                                                                                            | Production | Plant connection failed |
| SAK_1 | Based on pvsyst report, what is the rated power of this plant and array configuration                                                                                         | Production | Plant connection failed |
| SAK_1 | Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2                                                                                                  | Production | Plant connection failed |
| SAK_1 | Please plot last 3 days hourly performance ratio for my Strings for inverter 2                                                                                                | Production | Plant connection failed |
| SAK_1 | Please plot the last 4 days hourly performance ratio for all my inverters                                                                                                     | Production | Plant connection failed |
| SAK_1 | Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3                                                                | Production | Plant connection failed |
| SAK_1 | Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test) | Production | Plant connection failed |
| SAK_1 | How can I shut down my inverters?                                                                                                                                             | Production | Plant connection failed |

| Plant | Question                                                                                                                             | Env        | Skip reason             |
|-------|--------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------|
| SAK_1 | I have a COMMUNICATION_FAULT what do I do to my inverters?                                                                           | Production | Plant connection failed |
| SAK_1 | When was the last time the BOM data showed up                                                                                        | Production | Plant connection failed |
| SAK_1 | Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well      | Production | Plant connection failed |
| SAK_1 | Was there any 0 production days in the last 7 days?                                                                                  | Production | Plant connection failed |
| SAK_1 | Please change to dark mode                                                                                                           | Production | Plant connection failed |
| SAK_1 | Pull up the latest cleaning log                                                                                                      | Production | Plant connection failed |
| SAK_1 | Jim cleaned the pyranometer today, can you add it to the cleaning log?                                                               | Production | Plant connection failed |
| SAK_1 | Please generate December 2025 performance report                                                                                     | Production | Plant connection failed |
| SAK_1 | How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction? | Production | Plant connection failed |
| SAK_1 | How did my plant perform yesterday?                                                                                                  | Production | Plant connection failed |
| SAK_1 | Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt   | Production | Plant connection failed |
| SAK_1 | Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times                | Production | Plant connection failed |
| SAK_1 | Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well      | Production | Plant connection failed |